

Diagnostic Validation of a Portable Automotive RGB Lighting Platform Using the TLC6C5724-Q1 Automotive LED Driver

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I. INTRODUCTION

This report covers the diagnostic tests carried out on the TLC6C5724-Q1 platform. The tests were run with the ESP32 controller. Both healthy and fault cases were checked. Results were shown on the local TFT and on Node-RED.

The tested functions were ERR status, device status, IREF open and short, channel scan, LED open, LED short, output short-to-ground, APS, LOD/LSD self-test, NEG/GCLK test, and error clear.

II. SYSTEM UNDER TEST

The setup uses an ESP32, a TLC6C5724-Q1 LED driver, a TFT, push buttons, and a laptop. The ESP32 controls the TLC through the serial interface. It also provides GCLK and BLANK.

The TLC ERR pin is connected to the ESP32. ERR is active low. An external pull-up is used.

The grayscale mode is 12-bit. GCLK is about 1 MHz. The firmware can test all 24 output channels.

The laptop runs Mosquitto and Node-RED. MQTT carries commands and status data between the ESP32 and the dashboard.

The TLC6C5724-Q1 is a 24-channel constant-current RGB LED driver. It includes LED-open, LED-short, output short-to-ground, APS, thermal, IREF, NEG, and LOD/LSD test functions [1].

III. DIAGNOSTIC DATA PATH

The TLC returns diagnostic data through SID. SID contains channel fault bits and device status bits [1].

The ESP32 reads SID through SDO. The firmware then checks the required bits. The result is shown on the TFT and Node-RED.

Data path: Fault condition → TLC diagnostic logic → SID or ERR → ESP32 → TFT / MQTT → Node-RED.

TABLE I: Diagnostic fields used in the project

Field	Use
LOD1 / LOD2	LED open and output short-to-ground
LSD1 / LSD2	LED short
APS register + APS_FLAG	Adjacent-pin short result
TEF / PTW	Thermal status
ISF / IOF	IREF short and open status
LOD_LSD_FLAG	LOD/LSD self-test result
NEG1 / NEG2	NEG/GCLK test

TABLE II: Reference channel-fault patterns

Condition	LOD1	LOD2	LSD1	LSD2	Result
Healthy	0	0	0	1	OK
LED open	1	0	0	1	OPEN
LED short	0	0	1	1	LED SHORT
Output short-to-GND	1	1	0	0	OUT-GND

IV. TEST METHOD

The same basic test flow was used for the physical fault tests.

- 1) Start with the system in the healthy state.
- 2) Record the normal result.
- 3) Power off before changing the test wiring.
- 4) Apply the test condition.
- 5) Run the related diagnostic.
- 6) Check the result.
- 7) Remove the test condition.
- 8) Restore the normal wiring.
- 9) Clear stored errors when needed.
- 10) Check the healthy state again.

V. DIAGNOSTIC TESTS

A. Reference Fault Patterns

Table II shows the bit patterns used by the channel scan. These patterns match the TLC diagnostic data [1].

B. ERR Monitor

Test goal. Check the ERR signal in normal and fault states.

How it works. ERR is active low and open drain. The ESP32 reads the pin and sends the state to Node-RED [1].

Test steps.

- 1) Start with the normal setup.

TABLE III: Global SID status fields

Status	SID bit(s)	Normal	Fault
TEF	215	0	1
PTW	214	0	1
APS_FLAG	213..211	011	110
ISF	210	0	1
IOF	209	0	1
LOD_LSD_FLAG	208..206	011	110
NEG1 / NEG2	205 / 204	0 / 0	Toggle

TABLE IV: Special diagnostic commands

Command	Code	Use
ERROR Clear	0xA53	Clears stored error and APS status
APS check	0x53A	Runs the APS test
LOD/LSD self-test	0x535	Checks the LOD/LSD circuits
Negate Bit Toggle	0x55A	Toggles NEG bits

- 2) Check that ERR is high.
- 3) Run Device Status.
- 4) Apply an IREF fault.
- 5) Check that ERR goes low.
- 6) Remove the fault.
- 7) Clear the error.
- 8) Check that ERR returns high.

Expected. ERR is HIGH in the healthy state. ERR is LOW when the fault is active.

Result. ERR changed as expected.

Device overview	
UPTIME	00:06:08
ERR	OK
DEVICE	ONLINE
WIFI RSSI	-45 dBm
IP ADDRESS	192.168.0.247

(a) ERR = OK

Diagnostic Results	
DEVICE STATUS	OK
THERMAL ERROR	OK
THERMAL WARNING	OK
IREF OPEN	OK
IREF SHORT	OK
APS TEST	FAIL
APS RESULT	CS
LOD/LSD SELF TEST	PASS
NEGLOGX TEST	PASS
ERROR CLEAR	CLEARED
Diagnostic Controls	
CLEAR ERROR	NEGLOGX TEST
LOD/LSD TEST	APS TEST
SCAN CHANNELS	DEVICE STATUS
Device overview	
UPTIME	00:06:10
ERR	FAULT ACTIVE
DEVICE	ONLINE
WIFI RSSI	-44 dBm
IP ADDRESS	192.168.0.247

(b) ERR active during a fault

Fig. 1: ERR status in healthy and fault states.

C. Device Status

Test goal. Check TEF, PTW, ISF, and IOF.

How it works. TEF is SID bit 215. PTW is bit 214. ISF is bit 210. IOF is bit 209 [1].

Test steps.

- 1) Run Device Status in the healthy state.
- 2) Record TEF, PTW, ISF, and IOF.

- 3) Run the IREF open test.
- 4) Run the IREF short test.
- 5) Check the matching status bit.
- 6) Restore the IREF path.
- 7) Clear the error.

Expected. TEF, PTW, ISF, and IOF are 0 in the healthy state.

Result. The healthy status was correct. The IREF tests also set the correct status bit. TEF and PTW stayed clear during normal operation.

Diagnostic Results	
DEVICE STATUS	OK
THERMAL ERROR	OK
THERMAL WARNING	OK
IREF OPEN	OK
IREF SHORT	OK
APS TEST	PASS
APS RESULT	NONE
LOD/LSD SELF TEST	PASS
NEGLOGX TEST	PASS
ERROR CLEAR	CLEARED
Diagnostic Controls	
CLEAR ERROR	NEGLOGX TEST
LOD/LSD TEST	APS TEST
SCAN CHANNELS	DEVICE STATUS
Device overview	
UPTIME	00:07:28
ERR	OK
DEVICE	ONLINE
WIFI RSSI	-46 dBm
IP ADDRESS	192.168.0.247

(a) Healthy Device Status

Diagnostic Results	
DEVICE STATUS	FAULT SHORT
THERMAL ERROR	OK
THERMAL WARNING	OK
IREF OPEN	OK
IREF SHORT	ACTIVE
APS TEST	FAIL
APS RESULT	NONE
LOD/LSD SELF TEST	PASS
NEGLOGX TEST	FAIL
ERROR CLEAR	STILL ACTIVE
Diagnostic Controls	
CLEAR ERROR	NEGLOGX TEST
LOD/LSD TEST	APS TEST
SCAN CHANNELS	DEVICE STATUS
Device overview	
UPTIME	00:03:41
ERR	FAULT ACTIVE
DEVICE	ONLINE
WIFI RSSI	-41 dBm
IP ADDRESS	192.168.0.247

(b) Device Status during IREF short

Fig. 2: Device status in healthy and IREF-fault states.

D. Sequential Channel Scan

Test goal. Check the output channels one at a time.

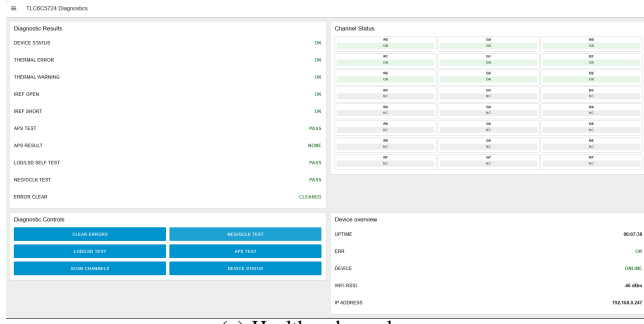
How it works. The firmware tests one output and reads LOD1, LOD2, LSD1, and LSD2. It then assigns the channel state.

Test steps.

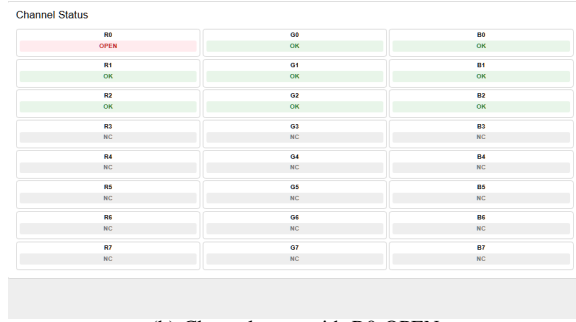
- 1) Run the scan in the healthy state.
- 2) Check that the active outputs show OK.
- 3) Apply one channel fault.
- 4) Run the scan again.
- 5) Check the affected channel.
- 6) Remove the fault.
- 7) Run the scan once more.

Expected. The fault should appear on the affected channel only.

Result. OPEN, LED SHORT, and OUT-GND were detected as separate states.

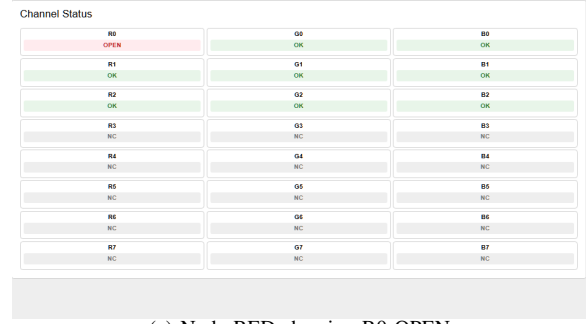


(a) Healthy channel scan

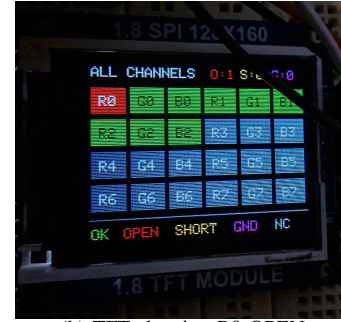


(b) Channel scan with R0 OPEN

Fig. 3: Channel scan in healthy and fault states.



(a) Node-RED showing R0 OPEN



(b) TFT showing R0 OPEN

Fig. 5: LED-open test result.

E. LED Open Test

Test goal. Check LED-open detection on one output.

How it works. The TLC checks the output voltage against the LOD threshold [1].

Test steps.

- 1) Power off the board.
- 2) Open the selected LED path.
- 3) Power the board.
- 4) Run Scan Channels.
- 5) Check Device Status.
- 6) Power off the board.
- 7) Restore the LED path.
- 8) Run the scan again.

Expected. The selected output should show OPEN. IREF OPEN should stay inactive.

Result. R0 showed OPEN. The other active outputs stayed normal.

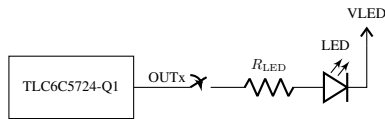


Fig. 4: Sample LED-open test circuit. The switch opens the selected LED path.

F. LED Short Test

Test goal. Check LED-short detection on one output.

How it works. The LSD circuit detects the LED-short condition [1].

Test steps.

- 1) Start with the healthy setup.
- 2) Apply the LED-short test condition.
- 3) Run Scan Channels.
- 4) Check the selected output.
- 5) Remove the test condition.
- 6) Run the scan again.

Expected. The selected output should show LED SHORT.

Result. B2 showed SHORT on the channel display.

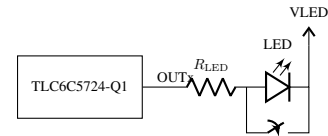
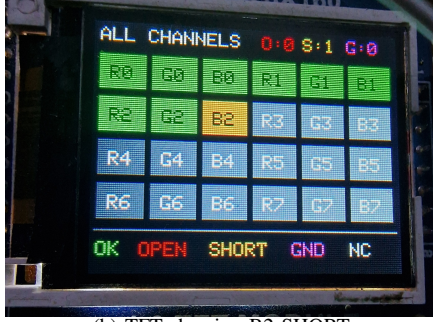


Fig. 6: Sample LED-short test circuit. The switch bypasses the selected LED.

Channel Status								
R0	OK	G0	OK	B0	OK			
R1	OK	G1	OK	B1	OK			
R2	OK	G2	OK	B2	SHORT			
R3	NC	G3	NC	B3	NC			
R4	NC	G4	NC	B4	NC			
R5	NC	G5	NC	B5	NC			
R6	NC	G6	NC	B6	NC			
R7	NC	G7	NC	B7	NC			

(a) Node-RED showing B2 SHORT

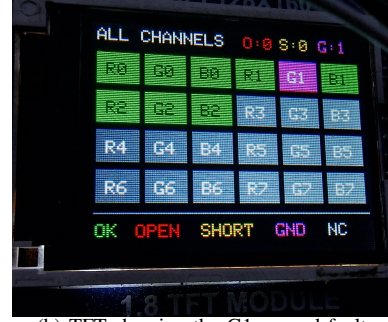


(b) TFT showing B2 SHORT

Fig. 7: LED-short test result.

Channel Status								
R0	OK	G0	OK	B0	OK			
R1	OK	G1	OUT GND	B1	OK			
R2	OK	G2	OK	B2	OK			
R3	NC	G3	NC	B3	NC			
R4	NC	G4	NC	B4	NC			
R5	NC	G5	NC	B5	NC			
R6	NC	G6	NC	B6	NC			
R7	NC	G7	NC	B7	NC			

(a) Node-RED showing G1 OUT GND



(b) TFT showing the G1 ground fault

Fig. 9: Output short-to-ground test result.

G. Output Short-to-Ground Test

Test goal. Check the output short-to-ground case.

How it works. The TLC uses a separate LOD/LSD pattern for this fault [1].

Test steps.

- 1) Power off the board.
- 2) Apply the output-to-ground test condition.
- 3) Power the board.
- 4) Run Scan Channels.
- 5) Check the selected output.
- 6) Remove the test condition.
- 7) Restore the normal setup.

Expected. The selected output should show OUT-GND.

Result. G1 showed OUT GND.

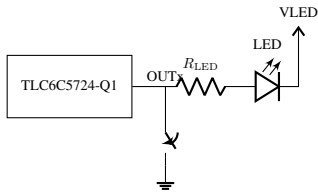


Fig. 8: Sample output-to-ground test circuit. The switch applies the test condition to the selected output.

H. Adjacent-Pin Short Test

Test goal. Check the APS function.

How it works. APS checks for a short between nearby output pins. APS_FLAG = 011b means pass. APS_FLAG = 110b means fail [1].

Test steps.

- 1) Run APS in the healthy state.
- 2) Record PASS.
- 3) Power off the board.
- 4) Apply the adjacent-pin test condition.
- 5) Power the board.
- 6) Run APS again.
- 7) Record the result.
- 8) Remove the test condition.
- 9) Clear the error.
- 10) Run APS again.

Expected. The healthy state should show PASS. The fault state should show FAIL and the affected output.

Result. APS passed in the healthy state. It failed during the test condition. G3 was shown as the affected result.

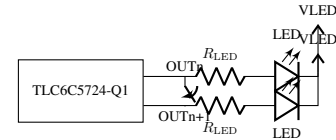
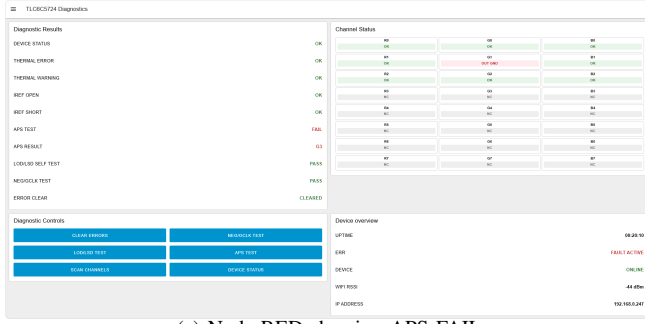
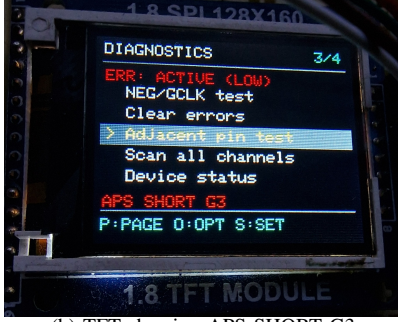


Fig. 10: Sample APS test circuit. The switch links two neighbouring output nodes.

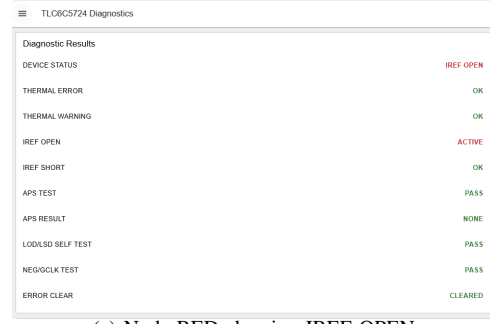


(a) Node-RED showing APS FAIL

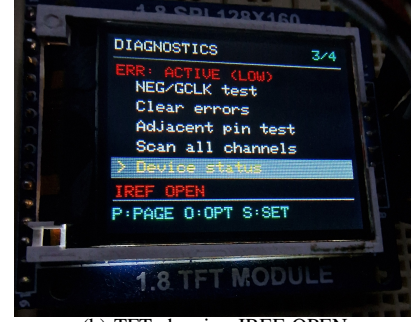


(b) TFT showing APS SHORT G3

Fig. 11: APS test result.



(a) Node-RED showing IREF OPEN



(b) TFT showing IREF OPEN

Fig. 13: IREF-open test result.

APS sequence: All outputs off → ERROR Clear 0xA53 → APS 0x53A → read SID → check APS result → restore normal output state.

I. IREF Open Test

Test goal. Check the IREF-open function.

How it works. IOF is set when the IREF current falls into the open range [1].

Test steps.

- 1) Record the healthy Device Status.
- 2) Power off the board.
- 3) Open the IREF path.
- 4) Power the board.
- 5) Run Device Status.
- 6) Check IOF and ERR.
- 7) Power off the board.
- 8) Restore the IREF path.
- 9) Power the board.
- 10) Clear the error.

Expected. IOF should be 1. ISF should be 0.

Result. Node-RED and the TFT showed IREF OPEN.



Fig. 12: Sample IREF-open test circuit. The switch opens the reference-resistor path.

J. IREF Short Test

Test goal. Check the IREF-short function.

How it works. ISF is set when the IREF current reaches the short range [1].

Test steps.

- 1) Start with the healthy IREF path.
- 2) Power off the board.
- 3) Apply the IREF-short test condition.
- 4) Power the board.
- 5) Run Device Status.
- 6) Check ISF and ERR.
- 7) Power off the board.
- 8) Restore the IREF path.
- 9) Power the board.
- 10) Clear the error.

Expected. ISF should be 1. IOF should be 0.

Result. Node-RED and the TFT showed IREF SHORT.

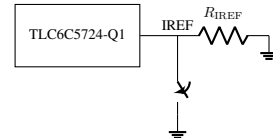
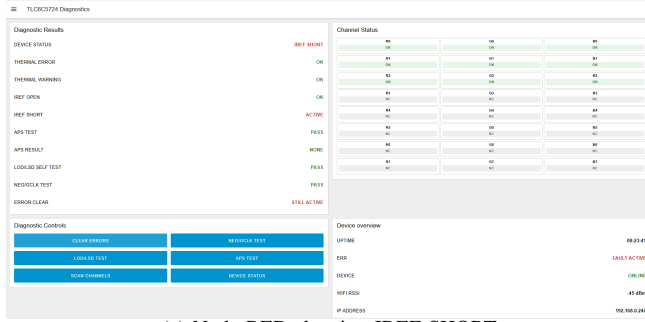
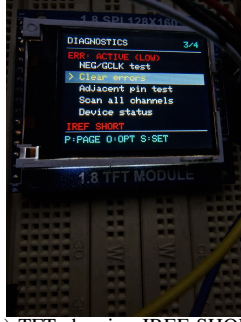


Fig. 14: Sample IREF-short test circuit. The switch applies the IREF short test condition.



(a) Node-RED showing IREF SHORT



(b) TFT showing IREF SHORT

Fig. 15: IREF-short test result.

K. LOD/LSD Self-Test

Test goal. Check the internal LED-open and LED-short detection circuits.

How it works. LOD_LSD_FLAG = 011b means pass. LOD_LSD_FLAG = 110b means fail [1].

Test steps.

- 1) Hold BLANK low.
- 2) Send 0x535.
- 3) Read SID.
- 4) Check bits 208..206.
- 5) Restore the normal LED state.

Expected. The test should return PASS.

Result. The LOD/LSD self-test returned PASS.

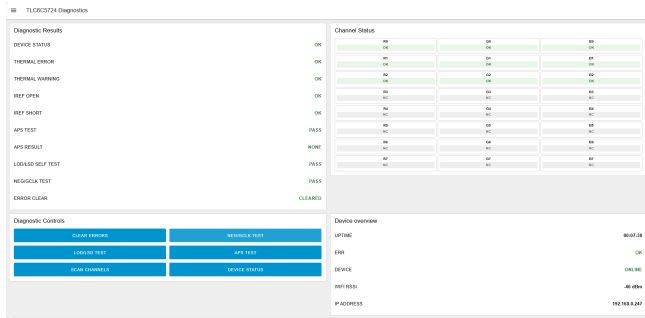


Fig. 16: LOD/LSD self-test result.

L. NEG/GCLK Test

Test goal. Check the NEG bits and GCLK-related register change.

How it works. NEG1 and NEG2 toggle after command 0x55A. The related LOD/LSD data changes with the toggle [1].

Test steps.

- 1) Apply the all-off diagnostic frame.
- 2) Read the first SID value.
- 3) Send 0x55A.
- 4) Read SID again.
- 5) Check NEG1 and NEG2.
- 6) Check the LOD/LSD change.
- 7) Send 0x55A again.
- 8) Check that the first state returns.

Expected. The NEG bits should toggle. The related data should change. The second command should restore the first state.

Result. The NEG/GCLK test returned PASS.

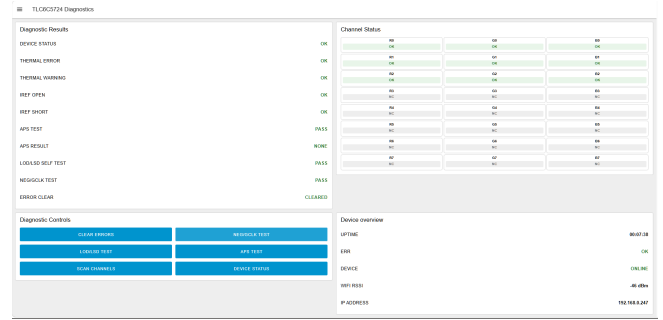


Fig. 17: NEG/GCLK test result.

M. Error Clear

Test goal. Clear stored diagnostic status before the next test.

How it works. ERROR Clear uses command 0xA53. It clears stored error and APS status [1].

Test steps.

- 1) Run a test that sets a stored status.
- 2) Restore the healthy setup.
- 3) Send Error Clear.
- 4) Run Device Status or APS again.
- 5) Check that the stored status is cleared.

Expected. ERROR CLEAR should show CLEARED.

Result. The stored status cleared correctly.

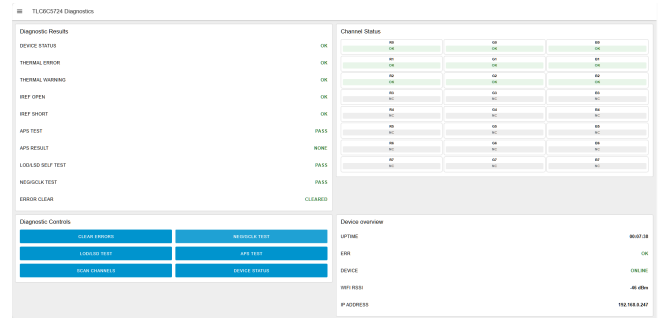


Fig. 18: Error-clear result.

TABLE V: Dashboard sections

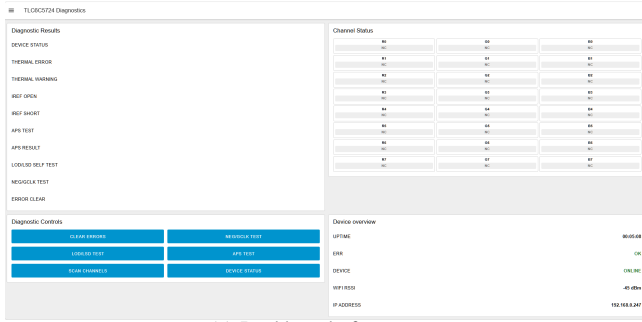
Section	Use
Diagnostic Results	Shows device and built-in test results
Channel Status	Shows OK, OPEN, SHORT, and OUT-GND
Diagnostic Controls	Runs the diagnostic tests
Device Overview	Shows device state, ERR, network data, and uptime

VI. NODE-RED DIAGNOSTIC DISPLAY

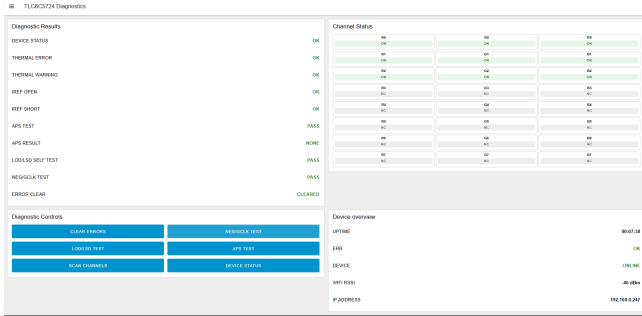
Node-RED shows the diagnostic results from the ESP32. It also provides the test buttons.

Command topic: `t1c6c5724/device01/command`

Status topics: `t1c6c5724/device01/#`



(a) Dashboard after startup



(b) Dashboard after the diagnostic checks

Fig. 19: Node-RED diagnostic display.

VII. VALIDATION SUMMARY

TABLE VI: Diagnostic Validation Summary

Test	Result	Status
ERR monitor	ERR changed between healthy and active fault states	Verified
Device status	TEF, PTW, ISF, and IOF were read correctly	Verified
LED open	Selected output showed OPEN	PASS
LED short	Selected output showed LED SHORT	PASS
Output short-to-GND	Selected output showed OUT-GND	PASS
APS	Adjacent-pin fault was detected	PASS
IREF open	IREF OPEN was detected	PASS
IREF short	IREF SHORT was detected	PASS
LOD/LSD self-test	Test returned PASS	PASS
NEG/GCLK	Test returned PASS	PASS
Error clear	Stored status was cleared	PASS
TEF/PTW	Both stayed clear in normal operation	Healthy

VIII. RESULTS

The channel scan detected LED open, LED short, and output short-to-ground as different fault types. APS also detected the adjacent-pin test condition.

The IREF tests worked as expected. IREF open set IOF. IREF short set ISF.

The LOD/LSD self-test returned PASS. The NEG/GCLK test also returned PASS. Error Clear removed the stored status after each test.

The same results were visible on the TFT and Node-RED.

IX. CONCLUSION

The TLC6C5724-Q1 diagnostics were tested with the ESP32 platform. The system detected the tested channel faults and device faults.

LED open, LED short, output short-to-ground, APS, IREF open, and IREF short were tested. The built-in LOD/LSD and NEG/GCLK checks also passed. Error Clear worked after the test conditions were removed.

The setup provides local and remote diagnostic testing for all 24 output channels.

REFERENCES

- [1] Texas Instruments, *TLC6C5724-Q1 Automotive 24-Channel, Full Diagnostics, Constant-Current RGB LED Driver*, rev. a ed., Texas Instruments, Aug. 2018.